

Problem

A voltage source supplies a signal of constant amplitude, from 0 to 40 kHz, to an RC low-pass filter. A load resistor, connected in parallel across the capacitor, experiences the maximum voltage at:

- (a) dc
- (b) 10 kHz
- (c) 20 kHz
- (d) 40 kHz

Step-by-step solution

Step 1 of 5

Consider the following data:

A voltage source supplies a signal of constant amplitude, and frequency varies from 0 to 40 kHz. Let us assume constant magnitude K .

Step 2 of 5

Consider a RC low-pass filter with a load resistor, connected in parallel across the capacitor:

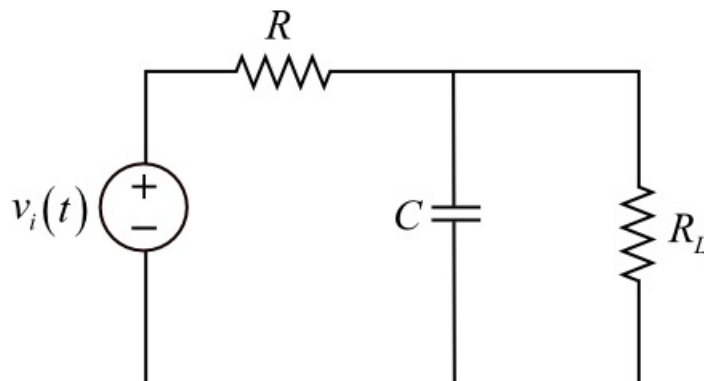


Figure 1

Step 3 of 5

We first covert the circuit to the frequency domain:

$$X_c = \frac{1}{j\omega C}$$

Find the impedance of the load resistor R_L in parallel with the capacitor C is $\mathbf{Z}$.

$$\begin{aligned}\mathbf{Z} &= R_L \parallel \frac{1}{j\omega C} \\ &= \frac{R_L \left(\frac{1}{j\omega C} \right)}{R_L + \frac{1}{j\omega C}} \\ &= \frac{\frac{R_L}{j\omega C}}{\frac{j\omega R_L C + 1}{j\omega C}} \\ &= \frac{R_L}{1 + j\omega R_L C}\end{aligned}$$

Step 4 of 5

Replace capacitor C and load resistor R_L with equivalent impedance $\mathbf{Z}$ as shown below.

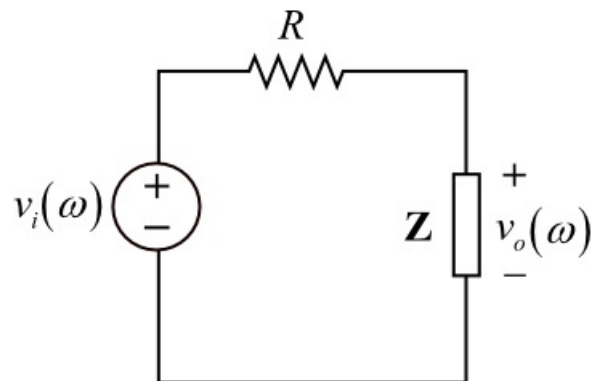


Figure 2

Step 5 of 5

Apply voltage division:

$$v_o(\omega) = \frac{Z}{R+Z} v_i(\omega)$$

Substitute $\frac{R_L}{1+j\omega R_L C}$ for Z in the above expression.

$$\begin{aligned} v_o(\omega) &= \frac{\left(\frac{R_L}{1+j\omega R_L C} \right)}{R + \left(\frac{R_L}{1+j\omega R_L C} \right)} v_i(\omega) \\ &= \frac{\frac{R_L}{1+j\omega R_L C}}{\frac{R(1+j\omega R_L C) + R_L}{1+j\omega R_L C}} v_i(\omega) \\ v_o(\omega) &= \frac{R_L}{R + R_L + j\omega R R_L C} v_i(\omega) \end{aligned}$$

Constant amplitude K for all the input frequencies ranging from 0 to 40 kHz. Thus substitute K for $v_i(\omega)$ in the above expression.

$$v_o(\omega) = \frac{K R_L}{R + R_L + j\omega R R_L C}$$

Find the output voltage at $\omega = 0$.

$$\begin{aligned} v_o(\omega) &= \frac{K R_L}{R + R_L + j(0) R R_L C} \\ &= \frac{K R_L}{R + R_L} \end{aligned}$$

At $\omega = 0$ the denominator term $j\omega R R_L C$ become zero. Remaining all the frequencies 10 kHz, 20 kHz and 40 kHz the denominator term $j\omega R R_L C$ increasing respectively which implies output voltage decreasing.

Thus for the RC low-pass filter produces maximum output voltage at $\omega = 0$ nothing but dc.

Therefore the correct option is .